

Lem var aleatorias indep iff sigma algebras indep

Lema 1. Sean X e Y dos variables aleatorias independientes en $(\Omega, \Sigma, \mathbb{P})$

$\iff \sigma(X)$ y $\sigma(Y)$ son σ -álgebras independientes.

Demostración: Sean $A_1 \in \sigma(X) \iff \exists E_1 \in \mathcal{B}(\mathbb{R}) : A_1 = X^{-1}(E_1)$. Análogamente, sea $A_2 \in \sigma(Y) \iff \exists E_2 \in \mathcal{B}(\mathbb{R}) : A_2 = Y^{-1}(E_2)$. Tomamos dichos E_1 y E_2 , entonces

$$\mathbb{P}(A_1 \cap A_2) = \mathbb{P}(\{X \in E_1\} \cap \{Y \in E_2\}) = \underset{\substack{\uparrow \\ \text{indep}}}{\mathbb{P}(X \in E_1) \cdot \mathbb{P}(Y \in E_2)} = \mathbb{P}(A_1) \cdot \mathbb{P}(A_2)$$

Como A_1 y A_2 eran arbitrarios, $\sigma(X)$ y $\sigma(Y)$ son independientes. ■

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